

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – V(Curiosity Question)

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ALGORITHM:

1. Start Visual Studio.
2. Create a new ASP.NET Core Web App (Model-View-Controller) project.
3. Install the required NuGet packages:
4. Microsoft.EntityFrameworkCore.SqlServer
5. Microsoft.EntityFrameworkCore.Tools
6. Microsoft.EntityFrameworkCore.Design
7. Microsoft.EntityFrameworkCore
8. Create a Product model with properties: Id, Name, Category, Price, and Quantity.
9. Create an ApplicationDbContext class and add **DbSet<Product>**.
10. Configure the SQL Server connection string in appsettings.json.
11. Register ApplicationDbContext in Program.cs using **AddDbContext()**.
12. Open Package Manager Console and execute:
13. **Add-Migration InitialCreate**
14. **Update-Database**
15. Right-click the Controllers folder and select Add → New Scaffolded Item.
16. Choose MVC Controller with views, using Entity Framework.
17. Select the Product model and ApplicationDbContext, then click Add.
18. Run the application.
19. Open the Products page.
20. Click Create New to add product details.
21. Save the product and verify it appears in the product list.
22. Use the Edit option to update product details.
23. Use the Details option to view complete product information.
24. Use the Delete option to remove a product.
25. Verify that all CRUD (Create, Read, Update, Delete) operations are working correctly.
26. Stop the application.

CODE:

(ProductsController.cs):

```
@model IEnumerable<Practical_5_CQ.Models.Product>

@{
    ViewData["Title"] = "Index";
}

<h1>Index</h1>

<p>
    <a asp-action="Create">Create New</a>
</p>
<table class="table">
    <thead>
        <tr>
            <th>
                @Html.DisplayNameFor(model => model.Name)
            </th>
            <th>
                @Html.DisplayNameFor(model => model.Category)
            </th>
            <th>
                @Html.DisplayNameFor(model => model.Price)
            </th>
            <th>
                @Html.DisplayNameFor(model => model.Quantity)
            </th>
        </tr>
    </thead>
    <tbody>
        @foreach (var item in Model) {
            <tr>
                <td>
                    @Html.DisplayFor(model => item.Name)
                </td>
                <td>
                    @Html.DisplayFor(model => item.Category)
                </td>
                <td>
                    @Html.DisplayFor(model => item.Price)
                </td>
                <td>
                    @Html.DisplayFor(model => item.Quantity)
                </td>
                <td>
                    <a asp-action="Edit" asp-route-id="@item.Id">Edit</a> |
                    <a asp-action="Details" asp-route-id="@item.Id">Details</a> |
                    <a asp-action="Delete" asp-route-id="@item.Id">Delete</a>
                </td>
            </tr>
        }
    </tbody>
</table>
```

(Program.cs):

```
using Microsoft.EntityFrameworkCore;
using Practical_5_CQ.Data;
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.
```

```

builder.Services.AddControllersWithViews();
builder.Services.AddDbContext<AppDbContext>(options =>
options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));

var app = builder.Build();

// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    // The default HSTS value is 30 days. You may want to change this for production
    scenarios, see https://aka.ms/aspnetcore-hsts.
    app.UseHsts();
}

app.UseHttpsRedirection();
app.UseRouting();

app.UseAuthorization();

app.MapStaticAssets();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}")
    .WithStaticAssets();

app.Run();

```

(appsetting.json):

```

{
  "ConnectionStrings": {
    "DefaultConnection":
    "Server=(localdb)\\MSSQLLocalDB;Database=ShoppingDB;Trusted_Connection=True;Trust
    ServerCertificate=True;",
    "AppDbContext":
    "Server=(localdb)\\mssqllocaldb;Database=AppDbContext;Trusted_Connection=True;Multipl
    eActiveResultSets=true"
  },
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}

```

(AppDbContext.cs):

```

using Microsoft.EntityFrameworkCore;
using Pratical_5_CQ.Models;
namespace Pratical_5_CQ.Data
{
    public class AppDbContext : DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
        {
        }

        public DbSet<Product> Products { get; set; }
    }
}

```

```
}
```

([Product.cs](#)):

```
namespace Pratical_5_CQ.Models
{
    public class Product
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Category { get; set; }
        public decimal Price { get; set; }
        public int Quantity { get; set; }
    }
}
```

Output:

Pratical_5_CQ Home Privacy

Index

[Create New](#)

Name	Category	Price	Quantity	
Laptop	Electronics	65000.00	5	Edit Details Delete
Mobile	Electronics	25000.00	10	Edit Details Delete
BackPack	Bag	1800.00	14	Edit Details Delete
Shoes	Footwear	3400.00	10	Edit Details Delete
Notebook	Stationery	120.00	100	Edit Details Delete
Pen Set	Stationery	250.00	200	Edit Details Delete

Pratical_5_CQ Home Privacy

Edit Product

Name

Category

Price

Quantity

[Back to List](#)

Pratical_5_CQ Home Privacy

Index

[Create New](#)

Name	Category	Price	Quantity	
Laptop	Electronics	65000.00	5	Edit Details Delete
Mobile	Electronics	25000.00	10	Edit Details Delete
BackPack	Bag	1800.00	14	Edit Details Delete
Shoes	Footwear	3400.00	10	Edit Details Delete
Notebook	Stationery	120.00	100	Edit Details Delete
Office Chair	Furniture	7500.00	80	Edit Details Delete

Delete

Are you sure you want to delete this?

Pratical_5_CQ.Models.Product

Name	Office Chair
-------------	--------------

Category	Furniture
-----------------	-----------

Price	7500.00
--------------	---------

Quantity	80
-----------------	----

[Delete](#) | [Back to List](#)